

FairShare Java Backend Roadmap

This roadmap guides you step by step to build the backend of FairShare (Trip Planning + Expense Tracking + Settlement System) in Java. It includes weekly milestones, exercises, and resources.

Week 1–2: Java Foundations (OOP, Collections)

- Learn Core Java (OOP, Collections, Exception Handling, File I/O).
- Exercises:
 - - Student class with grading system.
 - - Library System (Book, Member, Borrow) using ArrayList.
 - - Console Expense Tracker (HashMap).

Week 3: Database + JDBC

- Learn SQL (CREATE, INSERT, SELECT, UPDATE, DELETE).
- Practice JDBC (Connection, Statement, ResultSet).
- Exercises:
 - - Create User table and insert/fetch users via JDBC.
 - - Create Expense table and query total per user.

Week 4–5: Spring Boot & REST APIs

- Learn Spring Boot (Controllers, Services, Repositories).
- Build REST APIs for User, Expense, and Trip modules.
- Exercises:
 - - User Management API (CRUD).
 - - Expense API (Add, Fetch by user).
 - - Trip API (Create trip, Add members).

Week 6: Smart Settlement & Analytics

- Learn settlement algorithms (minimize transactions).
- Use SQL GROUP BY for analytics.
- Exercises:
 - - Implement settlement logic (who owes whom).
 - - Settlement API: /trips/{id}/settlement.
 - - Analytics API: /expenses/summary?userId=1.

Week 7: Security & Testing

- Learn Spring Security + JWT authentication.
- Write JUnit + Mockito tests.
- Exercises:
 - - Register/Login with JWT.
 - - Protect Expense APIs with JWT.
 - - Unit test SettlementService.

Week 8: Integration & Deployment

- Add Swagger for API docs.
- Deploy backend to Render/Heroku/Railway/AWS.
- Exercises:
 - - Swagger UI integration.
 - - Deploy working backend online.